

SARIMAX uses an exogenous regression structure with RMSE 1.063, MAE 0.915, and MAPE 8.808. The model uses an ARMA order of (1, 0, 1) with an intercept and the same exogenous variables used in the preprocessing step. LSTM uses one recurrent layer with dropout and early stopping, and its metrics are RMSE 0.076, MAE 0.076, and MAPE 1.249.

Residual diagnostics are reported for SARIMAX with a residual mean of -0.616 and a first-order autocorrelation of 0.128. The forecast plots are saved in the figures directory and the comparison is kept honest by reporting both the stronger and weaker model outputs rather than over-claiming performance. The main limitation is that the evaluation window is short and the LSTM setup is intentionally lightweight.

HCP Linkage Memo

Two HCP indicators were examined: FAO_CP_23012 and FAO_CP_23013. A simple regression was used to assess the association between those indicators and food inflation. The coefficient for FAO_CP_23012 is -0.402 with p-value 0.000, while the coefficient for FAO_CP_23013 is -0.486 with p-value 0.000. These results are interpreted as directional evidence rather than as a definitive causal claim, and the fitted relationship was then used to project simple forward paths for the indicators.